

Light-Weight Counter-Realities

Nicolas Romano, December 2024

It is the business of the future to be dangerous
—Alfred N. Whitehead, 1925

Introduction

Throughout the past 2000s, locative media has become an increasingly more integral part of daily life. As will be examined in this paper, several infrastructures have been developed including GPS, Bluetooth, WiFi, and RFID which interface with smart-phones to garner greater measurement of human positionality and movement. This essay explores two new types of location approaches that are on the rise including inertial based sensor method called pedestrian dead reckoning (PDR) and a camera based approach called Simultaneous Localization and Mapping (SLAM). It will be argued that these two technologies are approaching a convergence and that there are epistemological implications for human integration of these new forms of vision and embodiment. Three key areas will be explored including a historical background and competitive landscape existing during the development of foundational localization and mapping technologies. Second, a patent citation network analysis was performed to examine the strategy eminent within a leading algorithmic sensor based positioning and mapping company called Pointr. Third, an epistemological analysis will be taken with regard to SLAM and its consideration as a new type of vision and means of knowledge. Lastly, a hypothetical approach is encouraged towards the development of what I term as light-weight counter-realities as diverse forms of locally produced mixed reality experiences.

Infrastructural Battlegrounds of the Location Aware

In this section, we will follow works by media scholars who have utilized infrastructural analysis techniques to describe pre-eminent locative technologies including GPS, Bluetooth, WiFi, and RFID. Rowan Wilken describes this approach well in suggesting that *‘understanding specific instances of how these infrastructures were made gives us valuable insight into the often unseen workings of the larger mobile communication ecosystem, the corporate machinations that have shaped and continue to shape this ecosystem, and how it is that certain “keynote species” (Nardi & O’Day, 1999, p. 53), particularly Google, have come to dominate this ecosystem.’*¹ Tracing the strategic moves of companies in molding and building up particular technological infrastructures, as well as assessing the operativity and agentivity in everyday life of said technologies can provide a historical context, especially in the case of locative media, from which future ecosystems may unfold.

¹ Wilken, “Communication Infrastructures and the Contest over Location Positioning.”

First, we will look at Wilken's analysis of Skyhook Wireless's early contributions to mobile device locativity via WiFi infrastructures. Wilken accounts the ingenuity of Skyhook in envisioning a means to utilize the prolific emergence of WiFi routers in urban spaces as an infrastructure from which to pioneer location technology architectures. Skyhook utilized a wardriving approach by which they deployed a *'fleet of Wi-Fi scanning ("wardriving") cars... each car contained a laptop, window-mounted directional antennas, and technology that sent out "short blasts of radio waves, called probe requests, to detect nearby cell towers and Wi-Fi.'*² Among solutions to several early technical challenges in the development of mobile location technologies, this approach concretized as a foundational ingredient. The scanning systems enabled Skyhook to build up and continuously update a database which *'by 2009... include[d] 100 million wireless networks and 700,000 cell towers... handling 250 million requests per day.'*³ These databases served as a representation of the physical world from which triangulation algorithms between mobile devices, WiFi routers, and cell towers could be employed to establish a phone user's position with novel accuracy in the early 2000s. Wilken continues to argue that this model became the basis for Google's Street View and geolocation services API in the 2010s. He includes that Google Maps software deployed on mobile served as a collection method to *'gather MAC addresses and SSID data further enrich[ing] its database of existing Wi-Fi access points.'*⁴ Google extended upon Skyhook's model by also collecting *'Wi-Fi payload data [to] add contextual richness... by revealing user "habits, preferences, routes and routings" —information that is "critical to data-hungry economy".'*⁵ While Skyhook's location services were influential in early locative systems including Apple's 1st generation iPhone, Google questionably acquired crucial partnerships with phone providers Motorola and Samsung which shifted the market influence towards Google. In fact, Skyhook took legal action against Google claiming anticompetitive behavior and patent infringement, however, *'the two firms reached settlement just prior to trial, with Skyhook receiving US\$90 million in damages'*⁶ In retrospect, it is clear that this was a minimal expense for Google to establish dominance in the geolocation services arena at this critical stage in development. Subsequently, Skyhook attempted to remain relevant in a market dominated by giants by shifting their business model as a software SDK provider first to wearable technologies and then for mobile location analytics. Yet, these endeavors wouldn't re-establish their earlier market foothold.

While the locative infrastructure seemed to solidify into Google's hands, emerging arenas of bluetooth beacons and SaaS were viewed as frontiers of opportunity. Bluetooth Beacons came into cultural awareness in 2013 through Apple's iBeacon technology which reached places inaccessible to GPS and WiFi alone by establishing location tracking for indoor spaces. Nicholas and Shapiro described the potential of Bluetooth Beacons at the time as:

'Beacons can easily be installed out of sight. They require little maintenance and purport to improve tracking accuracy indoors and at short ranges. Whereas GPS and

² Wilken.

³ Wilken.

⁴ Wilken.

⁵ Wilken.

⁶ Wilken.

*WiFi-mapping pinpoint devices to the level of feet and meters, beacons promised precision in inches and centimeters*⁷

Yet, the arms race for Bluetooth Beacon technology was highly dependent upon the development of Bluetooth standards such as BLE 4.0. Standards served as the glue between the variety of mobile phone hardware and operating system stacks and the Beacon devices. Even though bluetooth infrastructure began to stabilize, *'marketers still faced interoperability challenges... [having] to write custom code to instruct apps to listen for beacons... [which] led to lock-in and dependencies that put off retail clients and constricted beacon networks' scalability.*⁸ For this reason, *'iOS and iBeacon became the most attractive combination for developers, while Apple maintained gatekeeping power through the App Store and iBeacon's certification program and licensing agreement'*⁹ While applications of Bluetooth beacons continue in retail spaces, the infrastructure did not reach its initial hype. Nicholas and Shapiro suggest that the infrastructure lacked scalability due to expensive physical hardware installation, there were exclusivity restrictions placed between hardware and software platforms, and the monetization potential of proximity marketing proved not as effective as anticipated. Further, companies including Foursquare found success in avoiding the physical infrastructural challenges of Beacons by building their *'Pilgrim SDK, which integrates GPS, WiFi, Bluetooth, cell tower, and sensor data to reconstruct consumer mobility'*¹⁰ Wilken and Frith also discuss Foursquares role as a location SaaS provider in noting that their Pilgrim SDK and Places API were used to power Yahoo's Flickr, Evernote, Twitter's Vine, early versions of Google's Waze, Facebook's Instagram, Mapbox, Uber, Snap, and Microsoft among others.¹¹

Despite the broad attention placed on locative infrastructures centered around the tracking of mobile phones, Frith urges not to overlook the widespread use of RFID as an infrastructure of locative intelligence. As there are substantially more RFID chips in circulation than mobile phones, RFID forms a network of *'invisible infrastructures that operate in the background of our interactions with space'*¹². Further, in addressing the effectiveness of RFID as operating in conjunction with Big Data cultures, Frith includes Katherine Hayles' remark that *'if relational databases are the brains of the system, RFID tags are the legs'*¹³ RFIDs can be embedded into most products such as cars, phones, and products on shelves so in the realm of Big Data, there emerges value in *'finding patterns in the randomness of large bodies of data'*¹⁴ Hayles identifies this as an embodiment shift whereby technologies *'used to track...specific configurations of bodies,... matter more than any individual embodiment.'*¹⁵ As an invisible technology compatible with invisible analytical systems, RFID signals us to question the agency of everyday objects as part of larger governance and surveillance architectures.

⁷ Nicholas and Shapiro, "Failed Hybrids."

⁸ Nicholas and Shapiro.

⁹ Nicholas and Shapiro.

¹⁰ Nicholas and Shapiro.

¹¹ Frith and Wilken, "Social Shaping of Mobile Geomedia Services."

¹² Frith, "Communicating behind the Scenes."

¹³ Frith.

¹⁴ Frith.

¹⁵ Frith.

Through this historical account of successive locative technologies with close attention to the infrastructural evolutions and operativity, it is revealed that successful location models often stick, but can become hybridized to meet new needs. Secondly, there is a trend where locative technologies became smaller from cell tower, to WiFi router, to Beacon, to RFID chip, and then hybrids amongst those. This is in alignment with the tendency for technological infrastructures to become invisible as they stabilize and form codependent relations with humans. As a key informational source in these analyses is the utilization of patent documentation as a means of understanding technological change. Wilken describes patents as:

Valuable instruments for seeking to understand how specific technical problems have been approached as sets of abstract relations and formal operations, and for identifying which aspects of this problem-solving process legal protection is being sought for.... patents are viewed and discussed not as technical blueprints or patterns, but, following Alain Pottage and Brad Sherman (2010), as “recipes” (p. 71) whose ingredients are “synergistic articulations” (p. 73) of old and new elements that seek, in combination, to “precipitate a particular effect” (p. 71) or achieve a certain outcome.¹⁶

In an effort to better understand the nature of emerging machine learning and sensor based infrastructures of mapping and localization, the next section will detail a patent network analysis with regard to the company Pointr.

Patent Networks

In the last section, we moved through several infrastructures of localization and mapping. Yet, this research project aimed to better understand the state of algorithmic ‘infrastructureless’ solutions to localization. These solutions rely on machine learning algorithms to interpret signals from various sensors on board of mobile phones. Solutions which involve use of the camera are often categorized under simultaneous localization and mapping (SLAM). Solutions which rely on MEMS sensors onboard of phones such as gyroscopes and accelerometers are typically categorized under Pedestrian Dead Reckoning. In this section, we will discuss a patent citation network analysis performed on an industry leading indoor mapping company called Pointr, who utilizes Pedestrian Dead Reckoning techniques in their customer solutions.

As discussed, Bluetooth Beacon technologies were coveted for their potential to enable positioning in indoor spaces, however, they did not garner as much long term stability as a solution due to difficulty in scalability of physical infrastructure. Pointr represents an established company offering a variety of localization solutions in the retail, events, hospitality, aviation, healthcare, and workplace spaces. Their solutions include indoor mapping & positioning & geofencing, analytics, geofencing, and software tools such as Pointr Cloud and Express. While they employ a variety of technologies discussed earlier in providing solutions for customers, their patented Deep Location tech is what sets them apart. Pointr’s Deep Location uses *‘sensor fusion techniques and machine-learning algorithms in order to deliver accuracy to within one*

¹⁶ Wilken, “Communication Infrastructures and the Contest over Location Positioning.”

meter.¹⁷ Naturally, I was more curious to understand how this technology worked which led me to their patents page.¹⁸

While navigating through their various patents, I realized that the patent citations section contained a rich web of information connecting Pointr to infrastructures introduced by other companies and research groups. For this reason, I gained a curiosity for the method of patent citation network graph analysis. In looking to a paper '*On a Patent Analysis Method for Technological Convergence*'¹⁹, an empirical method was proposed to measure technological trends and dynamics based on their patents. They suggest the following reasons for why patent information can be useful here:

First, a patent citation is defined as the frequency that a patent is cited in subsequent patents, which reflects the impact of its technological innovation and the pervasiveness of its technological information [22]. If citation indices such as citing-cited intensity and linkage are measured between technologies, patterns of technological innovation and knowledge flows can be identified [23]. Consequently, citation analysis has long been applied to understand linkages between industries or technologies, and further knowledge flows among them are investigated to identify the trends of knowledge or technology convergence. Second, a patent classification refers to a way the examiners of a patent office arranges patent documents according to the technical features of inventions. Since the same document may be classified in several classes, the co-classification information can be used to identify the relationships between technologies. If more and more patents are classified both by two classes, technologies in the two classes may have more in common and are getting closer to each other.²⁰

In the case of Pointr and as a proxy the use of algorithmic location techniques, the idea of convergence is to assess how this technology is being adopted and spread into various commercial sectors. While this paper helped to provide a framework for patent citation analysis, it didn't provide much methodology on how I could tailor this approach towards Pointr. A second paper entitled '*A Network Analysis Approach to Intellectual Property Research*'²¹ introduced *network analysis (NA) as an approach to patent citation analysis. This form of analysis 'offers several basic measures that describe the network structure and allow identification of important nodes. For instance, centrality measures allow to identify prominent nodes in the network; whereas, the density indicators give a synthetic measure of the degree to which the nodes are (or are not) a network's measures.'*²² Thus, a visual representation of Pointr's patent network could be derived, which by design of the network analysis graph, would indicate important nodes (citations), and reveal characteristics of the network such as centrality and clustered regions. In the next section, it will be described how a network graph was created for Pointr's patents.

¹⁷ "Indoor Positioning System | Pointr Deep Location®."

¹⁸ "Patents | Pointr."

¹⁹ Kim and Kim, "On A Patent Analysis Method for Technological Convergence."

²⁰ Kim and Kim.

²¹ Bekkers and Martinelli, "A Network Analysis Approach to Intellectual Property Research."

²² Bekkers and Martinelli.

Crawl, Configure, & Construct

While there are several graphical network analysis softwares available such as UCINET, Gephi, and Cytoscape among others, I took an approach to work at the programming language level as opposed to end user applications. Specifically, I made use of Javascript libraries called '3d-force-graph'²³ and 'd3-force-3d'²⁴ which were built upon a popular 3D Javascript library called Three.js.²⁵ I used these libraries because they offered a simple and highly customizable means to generate force-directed graphs of networks. A force-directed graph requires a list of nodes and links between nodes. In our case a node is a single citation and a link is created when one node cites another node. The complexity of connections between nodes in the network is then visualized in a physics simulation whereby each node is treated like an electric charge with attractive and repulsive forces to the other nodes. The final state is a static equilibrium which serves as a visual representation of the network. These 3D graphics libraries were particularly useful because they allow a viewer to click and drag to view different parts of the network in 3D space as well as zoom in or out through mouse wheel scrolling.

Next, it was equally important to create a means to gather all of the patent and citation data. While patent's data are publicly available, many services to gather data in bulk are placed behind a paywall such as SerpApi's Google Patents API²⁶ meaning that they give you 100 queries per month for free, but offer a variety of paid plans to up the number of queries to the 10,000s level. Knowing that this would be a limiting factor for the research, I opted to build a web crawling script²⁷ to gather the necessary data. This script was built in Python using the Requests²⁸ and BeautifulSoup²⁹ libraries. As opposed to web crawling approaches which emulate an active browser window such as Selenium³⁰. I preferred the requests approach which would generate an HTTP request to a citation's URL and return all the HTML information from the website³¹. From there, it was a matter of trial and error as well as reference to existing web-scrappers³² to locate and extract specific citation data. The data I gathered for each citation included its ID number, title, current assignee/company, priority date, classifications, as well as forward and backwards citations. Forward citations indicate the 'Cited By' table on Google's interface which means that this patent was cited by the following list of other patents. The Backward citations indicate that this current patent's design was influenced by the following list

²³ Asturiano, "Vasturiano/3d-Force-Graph."

²⁴ "Vasturiano/D3-Force-3d: Force-Directed Graph Layout in 1D, 2D or 3D Using Velocity Verlet Integration."

²⁵ "Mrdoob/Three.js: JavaScript 3D Library."

²⁶ SerpApi, "Google Patents API."

²⁷ Romano, "Pointr-Citation-Research/Pedestrian_dead_reckoning.Py at Main · Dr-Sax/Pointr-Citation-Research."

²⁸ "Requests: HTTP for Humans™ — Requests 2.32.3 Documentation."

²⁹ "Beautiful Soup Documentation — Beautiful Soup 4.12.0 Documentation."

³⁰ "Selenium with Python — Selenium Python Bindings 2 Documentation."

³¹ "Pointr-Citation-Research/Raw_patent_data/Example_google_patent_html_response.Html at Main · Dr-Sax/Pointr-Citation-Research."

³² Stevens, "RyanIstevens/Google_patent_scraper."

of existing patents. The BeautifulSoup library allowed me to locate the data corresponding to specific HTML tag names which serve as a proxy for what is visible to a user on the Google Patents site. The functional end result of this crawler script was the ability to enter a list of patents and export a JSON file³³ containing all of the patent data in an organized fashion.

The first JSON file was comprehensive of all patents located within Pointr's Deep Location Platform patent table³⁴ which served as a representation for their algorithmic localization technologies including geofencing, location analysis, blue-dot positioning, digital maps, and wayfinding. Next, the form of this JSON file had to be configured for compatibility with how the force-directed graph engine read in nodes and links. A JSON converter script³⁵ was written to succinctly arrange the node and link information into the proper format.³⁶ Additional functionality was added to this script to compute a size value which made node's visualized size to be proportional to the number of forward citations it had. Secondly, for many of the companies with influential patents, a link to their logo was collected so that it may be rendered as a place holder of the node. There were several hundred different nodes and companies, and automated retrieval of company logos was not as simple as hoped, so I opted to manually collect via company Facebook and LinkedIn pages. Nodes without logos are visualized as yellow squares.

Lastly, the force-directed graph³⁷ was stylized to enhance its effectiveness as a visual aid. A repulsive force was increased between nodes to avoid clusters of nodes to overlap blocking a clear visual. Second, a clicking feature was added where the user can click on a node's image tile and it will popup that patent's website in a new tab where you can view more information. Lastly, a hover feature was added where hovering the mouse pointer over a node's image tile will render a small description including the id, title, date, and company name on screen.

Graph Analysis

In analyzing the network graph, it is clear that several regions or clusters emerge. One cluster is centered around Pointr's patent US11514633B1 '*Systems and methods for automating conversion of drawings to indoor maps and plans*' and includes patents from companies like iRobot, BeamUp, Next, Geo Comm, and XLab among others. This cluster represents techniques for automatic conversion of 2d aerial images, drawn maps, or floor plans into 3D CAD and text databases which can serve as floor maps of buildings including furniture walls and any other large obstacles in a building. The next cluster is centered off of Pointr's patent US10378907 '*Methods and systems for determining vertical location in enclosed spaces*' and includes companies such as Ricoh, Rivada, Texas Instruments, Here Global, Qualcomm, Apple, NextNav, and Huawei. This cluster is all about how handheld device data such as change in barometric pressure can be used in conjunction with beacon data to calculate the vertical

³³ "Pointr-Citation-Research/Raw_patent_data/Pointr_patent_tree_l1.Json at Main · Dr-Sax/Pointr-Citation-Research."

³⁴ "Patents | Pointr."

³⁵ "Pointr-Citation-Research/Json_convert.Py at Main · Dr-Sax/Pointr-Citation-Research."

³⁶ "Pointr-Citation-Research/Formatted_patent_data/L1_l2_mix_tree_fmt.Json at Main · Dr-Sax/Pointr-Citation-Research."

³⁷ "Pointr-Citation-Research/Dynamic_graph.Html at Main · Dr-Sax/Pointr-Citation-Research."

position or floor that a handheld device is located within. The next cluster is centered around Pointr's patents US11169280 '*Systems and methods for direction estimation in indoor and outdoor locations*' and US10324197 '*Systems and methods for estimating initial heading at start-up of navigation*' which includes research groups like China University of Geosciences and Shanghai Jiaotong University as well as technology companies Apple, Honeywell, Qualcomm, Cubic, TDK, Agjunction, and TRX Systems. This cluster is dedicated towards techniques in locating devices by assessing their inertial motion including stop and start, and limiting position uncertainty. Most notably here is Apple's 2014 patent US20150350849A1 '*Location Determination Using Dual Statistical Filters*' which appears to be the first time Kalman Filtering was used as a machine learning technique to assess a phone's motion via sensor data. Lastly, I'd point out a large cluster which is isolated from the others centered around 3 Pointr Patents related to '*Systems and methods for location enabled search and secure authentication*' which includes many automotive patents including ones from Uber, Lyft, Ford, GM, BMW, Honda, Toyota, and Zoox as well as other tech companies such as Google Intel, T-Mobile, Blackberry, and Operr. This cluster is mainly oriented towards integrations between localization and safe guidance of a device user to their requested ride sharing service. Zoox, Uber, and Lyft among others had interesting patents pertaining to augmented reality interfaces which would assist a device user in initiating rides and safely organizing a pickup. This indicated the access of a user's camera to perform SLAM modelling of the environment.

Through this visual analysis, it is clear that Pointr is not alone in developing this technology as there are hundreds of other companies and patents composed in their citation map. From a strategy and problem solving perspective, it appears that Pointr is targeting not only solutions which can effectively map indoor spaces and provide real-time positioning and navigational experiences for users, but also trying to integrate with the automotive industry in terms of ride sharing services and autonomous vehicles. This is indicative of a broader continuity by which a device user is integrated within a smart and precise navigational space when inside a Pointr building, and is seamlessly transferred to a transportation service which further captures their movement. As autonomous vehicle companies are increasingly incorporating SLAM technologies to support real-time autonomous driving as well as in AR interfaces to assist users in navigating, it is possible that pedestrian dead reckoning methods including Kalman Filtering are undergoing a convergence with SLAM camera based navigation systems. While this conclusion is based upon inference into a single company's patents and strategic direction, it could serve as a seed for a larger scale investigation into the question. In the next section, we will focus closer on an epistemological view of SLAM technologies.

Algorithmic Machines of the Invisible

As discussed, a new form of algorithmic sight and wayfinding called SLAM is emerging as a dominant technology in autonomous motiles such as in assisted driving and robotic manufacturing. It will be contended that this technology represents an alternative form of vision with potential to alter modes of human perception. In their analysis of new generations of geomeia technologies, Kanderske and Thielmann mark SLAM as an epistemological shift in perception as it is not based upon reference to 2D ground truths such as traditional hand-drawn

maps or GPS based navigation systems. On the contrary, SLAM systems engage in a form of self navigation through networked space and time. They write that *'a voyage cannot be considered as an isolated event. It has to be understood as the $n + 1$ th link in a long chain, where n is the tally of preceding exploratory enterprises, the experiences of which the current explorers can draw on.'*³⁸ In other words, a locative media device equipped with SLAM is in a continuous process of generating a spatial representation of its environment with reference to its previous faulty observations. As robotic motiles attempt to map the environment they are in, there is an intractable chicken-egg dilemma by which sensor error leads to faulty representation of the robot's current position which perpetuates into an inaccurate mapping of the environment and feeds back into continued errors of the former. For SLAM devices:

*'Both problems have to be solved at the same time, simultaneously, while taking the uncertainty of the robot's sensory measurements—and the effects this uncertainty has on all future measurements—into account. It follows that the resulting model of the world cannot be deterministic (it cannot be computed with certainty from an initial state), but has to be probabilistic: For every given step of the process, there is a given probability that the measured state, in which both the map and the pose of the robot are compounded, is correct.'*³⁹

This unique emergent form of vision can be best characterized as a way of knowing in the present which is highly dependent upon the varieties of reference data gathered from past explorations.

With regard to humans, opting to use SLAM devices as wayfinding technologies, there remains possibility for alterations in visual perception practices. In building a phenomenology of mobile phone usage in pedestrian place-making, Wilken and Richardson describe device-assisted wayfinding as a form of haptic vision. These dynamics can be described as a practice that *'incorporates two simultaneous perspectives—a "grounded" perspective of their own bodily positioning in urban space, and a bird's-eye view of their position and movement on a map. Both of these perspectives are achieved via "adroit oscillation" between stickiness and distraction, and dynamic modes of "haptic vision" that involve eyes, hands, feet and mobile screen device.'*⁴⁰ This suggests that users of traditional GPS based mobile phones for urban navigation require bodily practices incorporating multimodal sensory comparisons from walking, visual reference to surroundings of buildings and streets, and tactile pinch and zoom of a bird's eye map to assist in navigation. It is with SLAM, that the agency of these embodied micro-gestures are passed over to sensors equipped with probabilistic algorithms to resolve locative uncertainties in environmental representation and navigation.

In this transfer of embodied navigational agency, it can be argued that the simplicities and new visualities afforded by SLAM based capitalist media technologies come at the price of algorithmic invisibility. Ozkul invokes Oscar Gandy's 'The Panoptic Sort'⁴¹ in describing this

³⁸ Kanderske and Thielmann, "Simultaneous Localization and Mapping and the Situateness of a New Generation of Geomedia Technologies."

³⁹ Kanderske and Thielmann.

⁴⁰ Richardson and Wilken, "Haptic Vision, Footwork, Place-Making."

⁴¹ Jr, *The Panoptic Sort*.

situation as the algorithmic fix which acts as an ‘all seeing eye’ in guiding the global capitalist system. Ozkul writes:

*These profiles along with the information about our mobile habits then feed into various mobility systems, creating a perpetual cycle of accumulation of mobile sensor and user data, sorting and processing those data, and creating personal profiles that we are eventually driven to match. This fixes who we are and whom we may become, creating predictable futures where individual or collective identities and placemaking practices are homogenized, the behaviors that are seen as risky or not conforming are fixed by preventing any type of mobilities. Hence, the future of societies governed by algorithmic systems are somehow made predictable.*⁴²

In other words, the sheer magnitude and precision of locative pedestrian data made available to the cultural industries, and increased pedestrian reliance on platformized hermeneutic visualization tools as a mode of seeing and making decisions has risks. It signals the potential for a guiding hand to shape development of a person’s choices and individuality through algorithmic opaqueness.

Egliston and Carter invoke a similar proposition with regard to mixed reality (MR) technologies which often incorporate SLAM visualities. They engage with Bernard Stiegler’s philosophy of technics in warning that ‘*such visions of an MR future present perception as the outcome of a technological collapsing of future and past into present - rendering individuals as more profitable consumers, productive workers, or subjects to be disciplined by the state*’⁴³ In similarity, SLAM’s visual architecture situates humans in an altered spatio-temporal landscape by which algorithmic machines hold responsibility in comparing past visualities to form a representation of the present and assist in navigation of the future. Further, Stiegler is invoked by Azar, Cox, and Impett in suggesting that these visual practices have potential to de-link processes of seeing and knowledge formation. In describing algorithmically enabled visualities, they write:

*Although we see more, we are not given the instruments to understand the ways in which we see more—this being mainly a problem of property, for example in relation to our reliance on proprietary platforms and infrastructures, but also a problem related to the complexity of algorithmic networks, which in their operations often escape current epistemological frameworks.*⁴⁴

These warnings signal the importance for designers to ‘*articulate the kinds of futures we want from MR, and to start the conversations about how we might ‘care’ (as Stiegler would have it) for MR as an emerging part of our sociotechnical infrastructure.*’⁴⁵ For this to be a possibility, tools must be available for the average person to engage in the production of MR outside of the homogenized visualities made available by large tech platforms. The next section describes a hypothetical do-it-yourself (DIY) approach in building lightweight and shareable ‘*counter*’-reality experiences.

⁴² Ozkul, “The Algorithmic Fix.”

⁴³ Egliston and Carter, “The Interface of the Future.”

⁴⁴ Azar, Cox, and Impett, “Introduction.”

⁴⁵ Egliston and Carter, “The Interface of the Future.”

The DIY MR Tutorial

Now, while the SDK model in localization technology initiated by Foursquare's Pilgrim SDK is common and also included by Pointr in their Mobile SDK⁴⁶, it is evident that these algorithmic inertial sensor based pedestrian dead reckoning infrastructures will not be cheaply or freely available for the public for some time. While there have been several open source projects dedicated to PDR in the phone browser such as SmartPDR⁴⁷, the project hasn't been updated for 3 years indicating that support may be limited. Thus, for local, small-scale development, mixed reality implementations will be most functional and possible using SLAM or camera based approaches. In the desire for light-weight solutions and highly customizable solutions, we will only look to solutions which can run MR cross-platform (iOS and Android) in the browser as opposed to in app form. This eliminates large platforms such as Google's ARCore which need to be developed in heavy software development environments such as Adobe Aero or Unity and only outputs experiences in the form of apps.

While apps are a valid form of engaging users in an MR experience, and lend themselves to more efficient utilization of mobile device graphics cards, many softwares, including Pointr, have successfully developed localization systems which run in the phone browser. Browser hosting of MR programs is favorable because it is inline with the cultural language of link sharing and QR codes. Furthermore, techniques in Web Assembly (WASM) are gaining traction which *'enable web browsers to execute code at near-native speed by providing a compact binary format that is designed for fast decoding and execution.'*⁴⁸ As such, there are several open source SLAM projects incorporating WASM such as WASM-AR⁴⁹ and AlvaAR⁵⁰. The format of these projects can be suitable for rapid prototyping of new MR experiences.

However, to get even more lightweight and simplistic, an immersive web-app may be developed using a form of QR codes called ArUco⁵¹ coupled with a mobile implementation of computer vision library OpenCV⁵². This approach is also mentioned in Kanderske and Thielmann's paper in which they write:

*Several approaches make use of the barcode's spiritual successor, the QR code. These codes, placed around a room, can provide fixed landmarks that the robot or interface can optically track. They subsequently allow the rest of the space to be measured in relation to said landmarks*⁵³

To add more detail, the computer vision algorithm must be calibrated to parameters specific to the device's camera such as focal length. After calibration, the ArUco code can be recognized as indicative that a user has reached a specific region. Secondly, the ArUco code's pose

⁴⁶ "Getting Started - Pointr Docs."

⁴⁷ "Firecommit/React-Native-Smartpdr."

⁴⁸ "Leveraging WebAssembly for Augmented and Virtual Reality - FriendlyUsers Tech Blog."

⁴⁹ Lu, "EdwardLu2018/Wasm-Ar."

⁵⁰ "Alanross/AlvaAR: World Tracking for WebAR. A Javascript Library for Augmented Reality to Run SLAM in the Browser."

⁵¹ Mellado, "Jcmellado/Js-Aruco."

⁵² "OpenCV: Introduction to OpenCV.Js."

⁵³ Kanderske and Thielmann, "Simultaneous Localization and Mapping and the Situateness of a New Generation of Geomedia Technologies."

(distance away from the user and angle of orientation) can be estimated. This information can be used in a variety of potential ways such as the program triggering audio to play based on user position, angle, or distance. Secondly, this information can be integrated with a 3D graphics engine such as Three.js discussed earlier in which graphics may be layered upon the real world and interacted with through user touch or any number of sensor inputs.

The discussion of this hypothetical MR experience is to indicate the possibility for the average person to develop customizable MR applications which may be widely sharable and accessible. As mentioned previously, the importance of individually devised MR experience as opposed to ones produced by large platforms and heavy duty softwares is of increasing importance to help individuals recover visibility of infrastructures which are vanishing into the peripheries of awareness. Secondly, it gives individuals participatory agency in shaping the form, aesthetics, functionality, and diversity of MR experiences. To better describe layered experiences of this form, I drop the 'mixed' in mixed-reality in favor of 'counter' in counter-reality (CR).

Conclusion

By engaging with the historic developments of localization infrastructure, it was observed that much competition and desire for infrastructural control existed within the development of early GPS, Bluetooth, WiFi and RFID location techniques. Secondly, these infrastructures gradually became invisible to users as functionality improved, standards developed, and popular applications made real-time use of them. By engaging in a graphical network analysis of Pointr, it was suggested that the emergence of indoor localization and mapping through sensor focused approaches is converging with infrastructures of ride sharing and autonomous vehicles. In this way there may no longer be a divide between indoor and outdoor in terms of location tracking as previously segregated infrastructures interconnect. Lastly, the epistemological analysis of SLAM as a mode of vision suggested that we take care to retain awareness and visibility of the ways in which algorithmic localization may alter modes of embodiment and individuality with regard to capitalist cultural economies. The proposal of a hypothetical counter-reality experience is included as my own attempt to offer an approach to care in this scenario.

References

- “Alanross/AlvaAR: World Tracking for WebAR. A Javascript Library for Augmented Reality to Run SLAM in the Browser.” Accessed December 13, 2024.
<https://github.com/alanross/AlvaAR>.
- Asturiano, Vasco. “Vasturiano/3d-Force-Graph.” HTML, December 13, 2024.
<https://github.com/vasturiano/3d-force-graph>.
- Azar, M., G. Cox, and L. Impett. “Introduction: Ways of Machine Seeing.” *AI & Society: Journal of Knowledge, Culture and Communication* 36, no. 4 (February 20, 2021): 1093–1104.
- “Beautiful Soup Documentation — Beautiful Soup 4.12.0 Documentation.” Accessed December 13, 2024. <https://www.crummy.com/software/BeautifulSoup/bs4/doc/#>.
- Bekkers, Rudi, and Arianna Martinelli. “A Network Analysis Approach to Intellectual Property Research.” In *Handbook of Intellectual Property Research: Lenses, Methods, and Perspectives*, edited by Irene Calboli and Maria Lillà Montagnani, 0. Oxford University Press, 2021. <https://doi.org/10.1093/oso/9780198826743.003.0033>.
- Egliston, Ben, and Marcus Carter. “‘The Interface of the Future’: Mixed Reality, Intimate Data and Imagined Temporalities.” *Big Data & Society* 9, no. 1 (January 1, 2022): 20539517211063689. <https://doi.org/10.1177/20539517211063689>.
- “Firecommit/React-Native-Smartpdr.” JavaScript. 2021. Reprint, Firecommit, August 7, 2024.
<https://github.com/Firecommit/react-native-smartpdr>.
- Frith, Jordan. “Communicating behind the Scenes: A Primer on Radio Frequency Identification (RFID).” *Mobile Media & Communication* 3, no. 1 (January 2015): 91–105.
<https://doi.org/10.1177/2050157914554728>.
- Frith, Jordan, and Rowan Wilken. “Social Shaping of Mobile Geomedia Services: An Analysis of Yelp and Foursquare.” *Communication and the Public* 4, no. 2 (June 1, 2019): 133–49.
<https://doi.org/10.1177/2057047319850200>.
- “Getting Started - Pointr Docs.” Accessed December 13, 2024.
<https://docs-dev.pointr.tech/docs/8.x/Developer%20Portal/Getting%20Started/Pointr%20Mobile%20SDK/>.
- “Indoor Positioning System | Pointr Deep Location®.” Accessed December 12, 2024.
<https://www.pointr.tech/solutions/location-based-services/indoor-positioning>.
- Jr, Oscar H. Gandy. *The Panoptic Sort: A Political Economy of Personal Information*. Second Edition, New to this Edition:, Second Edition, New to this Edition: Oxford, New York: Oxford University Press, 2021.
- Kanderske, Max, and Tristan Thielmann. “Simultaneous Localization and Mapping and the Situateness of a New Generation of Geomedia Technologies.” *Communication and the Public* 4, no. 2 (June 1, 2019): 118–32. <https://doi.org/10.1177/2057047319851208>.
- Kim, Moon-Soo, and Chulhyun Kim. “On A Patent Analysis Method for Technological Convergence.” *Procedia - Social and Behavioral Sciences*, ASIA PACIFIC BUSINESS INNOVATION AND TECHNOLOGY MANAGEMENT SOCIETY, 40 (January 1, 2012): 657–63. <https://doi.org/10.1016/j.sbspro.2012.03.245>.
- “Leveraging WebAssembly for Augmented and Virtual Reality - FriendlyUsers Tech Blog.” Accessed December 13, 2024.
https://friendlyuser.github.io/posts/tech/2023/Leveraging_WebAssembly_for_Augmented_and_Virtual_Reality/.
- Lu, Edward. “EdwardLu2018/Wasm-Ar.” JavaScript, December 4, 2024.
<https://github.com/EdwardLu2018/wasm-ar>.
- Mellado, Juan. “Jcmellado/Js-Aruco.” JavaScript, October 29, 2024.
<https://github.com/jcmellado/js-aruco>.
- “Mrdoob/Three.js: JavaScript 3D Library.” Accessed December 13, 2024.
<https://github.com/mrdoob/three.js>.

Nicholas, Gabriel, and Aaron Shapiro. "Failed Hybrids: The Death and Life of Bluetooth Proximity Marketing." *Mobile Media & Communication* 9, no. 3 (September 1, 2021): 465–87. <https://doi.org/10.1177/2050157920975836>.

"OpenCV: Introduction to OpenCV.js." Accessed December 13, 2024. https://docs.opencv.org/4.x/df/df7/tutorial_js_table_of_contents_setup.html.

Özkul, Didem. "The Algorithmic Fix : Location Intelligence, Placemaking, and Predictable Futures." *Convergence. The International Journal of Research into New Media Technologies* 27, no. 3 (2021): 594–608.

"Patents | Pointr." Accessed December 12, 2024. <https://www.pointr.tech/technology/patents>.

"Pointr-Citation-Research/Dynamic_graph.Html at Main · Dr-Sax/Pointr-Citation-Research." Accessed December 13, 2024. https://github.com/dr-Sax/Pointr-Citation-Research/blob/main/dynamic_graph.html.

"Pointr-Citation-Research/Formatted_patent_data/L1_I2_mix_tree_fmt.Json at Main · Dr-Sax/Pointr-Citation-Research." Accessed December 13, 2024. https://github.com/dr-Sax/Pointr-Citation-Research/blob/main/formatted_patent_data/l1_i2_mix_tree_fmt.json.

"Pointr-Citation-Research/Json_convert.Py at Main · Dr-Sax/Pointr-Citation-Research." Accessed December 13, 2024. https://github.com/dr-Sax/Pointr-Citation-Research/blob/main/json_convert.py.

"Pointr-Citation-Research/Raw_patent_data/Example_google_patent_html_response.Html at Main · Dr-Sax/Pointr-Citation-Research." Accessed December 13, 2024. https://github.com/dr-Sax/Pointr-Citation-Research/blob/main/raw_patent_data/example_google_patent_html_response.html.

"Pointr-Citation-Research/Raw_patent_data/Pointr_patent_tree_l1.Json at Main · Dr-Sax/Pointr-Citation-Research." Accessed December 13, 2024. https://github.com/dr-Sax/Pointr-Citation-Research/blob/main/raw_patent_data/pointr_patent_tree_l1.json.

"Requests: HTTP for Humans™ — Requests 2.32.3 Documentation." Accessed December 13, 2024. <https://requests.readthedocs.io/en/latest/>.

Richardson, Ingrid, and Rowan Wilken. "Haptic Vision, Footwork, Place-Making: A Peripatetic Phenomenology of the Mobile Phone Pedestrian," January 1, 2009. <https://doi.org/10.25916/sut.26254085.v1>.

Romano, Nicolas. "Pointr-Citation-Research/Pedestrian_dead_reckoning.Py at Main · Dr-Sax/Pointr-Citation-Research." GitHub. Accessed December 13, 2024. https://github.com/dr-Sax/Pointr-Citation-Research/blob/main/pedestrian_dead_reckoning.py.

"Selenium with Python — Selenium Python Bindings 2 Documentation." Accessed December 13, 2024. <https://selenium-python.readthedocs.io/>.

SerpApi. "Google Patents API." SerpApi. Accessed December 13, 2024. <https://serpapi.com/google-patents-api>.

Stevens, Ryan. "Ryanlstevens/Google_patent_scraper." Python, December 11, 2024. https://github.com/ryanlstevens/google_patent_scraper.

"Vasturiano/D3-Force-3d: Force-Directed Graph Layout in 1D, 2D or 3D Using Velocity Verlet Integration." Accessed December 13, 2024. <https://github.com/vasturiano/d3-force-3d>.

Wilken, Rowan. "Communication Infrastructures and the Contest over Location Positioning." *Mobile Media & Communication* 7, no. 3 (September 1, 2019): 341–61. <https://doi.org/10.1177/2050157919847503>.